

Documentation

Introduction

The documentation talks about the data cleaning process for the datasets, why some choices were made, examples and reasoning behind them. It provides evidence of calculated fields used inside Tableau with a short explanation for what each calculated field does and talks about the joins that were performed to finalise the dashboard.

Physical Sales of Video Games Trending

VGA Dataset from- <https://zenodo.org/records/5898311>

Scraped from- <https://www.vgchartz.com>

Gaming Companies dataset from-

<https://www.kaggle.com/datasets/kkhandekar/large-video-game-companies/data>

Sourced from- <https://companiesmarketcap.com>

Data Cleaning

VGA Dataset	- Initially there were multiple name errors due to foreign games using foreign alphabet, those were checked online to see the correct spelling and replaced. - Some names had repeated values that could be found in a different column, so they were removed as they were redundant such as having “(US Sales)” which is represented by having a sales number in the “US Sales” column and similarly for having “(3DS)” which would be stated under ‘Platform’ column - Game titles do include the date like “Fifa 15” where the number represents the year, but the game itself came out in 2014, so the number is not removed as it is part of the name and doesn’t indicate the year. - There are repeating game titles however they were left in because some games were released on several platforms so the data they show differs. - Where the year was (blank) or “N/A” it was replaced for “Unknown” for completeness.
Gaming Companies Dataset	- The gaming companies’ dataset columns that were deemed to not be needed were removed: employee count, operating margin, PE ratio. - Companies that had values of 0 for most of the remaining columns were removed. - Discrepancies between the companies’ names between the two datasets were fixed so they matched. - Values were formatted so they were in USD

Year	Year fixed
N/A	Unknown
N/A	Unknown
N/A	Unknown
N/A	Unknown
N/A	Unknown
N/A	Unknown

Done by using a helper column and an IF statement.

- Found rounding errors in global sales calculations which represents the total of the several region sales.

NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales	Global_Sales fixed
41.49	29.02	3.77	8.46	82.74	82.74
29.08	3.58	6.81	0.77	40.24	40.24
15.85	12.88	3.79	3.31	35.82	35.83
15.75	11.01	3.28	2.96	33	33
11.27	8.89	10.22	1	31.37	31.38
23.2	2.26	4.22	0.58	30.26	30.26
11.38	9.23	6.5	2.9	30.01	30.01
14.03	9.2	2.93	2.85	29.02	29.01
14.59	7.06	4.7	2.26	28.62	28.61
26.93	0.63	0.28	0.47	28.31	28.31
9.07	11	1.93	2.75	24.76	24.75

Solved by using a helper column with a sum function on the region sales.

Examples of data cleaning

Name Miss-Spelling	- Wreckless: ThE YaKuza MisSiOns -> Wreckless: The Yakuza Missions - Pro Yaky? Spirits 4 -> Pro Yakyuu Spirits 4 - Kamaitachi no Yoru Ã— 3 -> Kamaitachi no Yoru x3 - PokÃ©mon -> Pokémon
Name duplicating values	- Cabela's Outdoor Adventures (2009) -> Cabela's Outdoor Adventures - Naruto: Ninja Destiny (US sales)-> Naruto: Ninja Destiny - International Superstar Soccer 2000 (JP weekly sales) -> International Superstar Soccer 2000 - The Legend of Korra (3DS) -> The Legend of Korra
Discrepancies between names	- Activision Blizzard -> Activision - Take 2 Interactive -> Take-Two Interactive - Sony -> Sony Computer Entertainment

Calculated fields

Dummy field- Information

```
// Creates an empty datapoint, used to change it into an icon for information in the dashboard  
d
```

EU % of Total

```
// Works out the % Sales for the EU Region  
SUM([EU Sales]) / ([Total Sales (All Regions)])
```

JP % of Total

```
// Works out the % Sales for the JP Region  
SUM([JP Sales]) / ([Total Sales (All Regions)])
```

NA % of Total

```
// Works out the % Sales for the North American (NA) Region  
SUM([NA Sales]) / ([Total Sales (All Regions)])
```

Other % of Total

```
// Works out the % Sales for the Other Regions  
SUM([Other Sales]) / ([Total Sales (All Regions)])
```

Region Sales

```
// Filters data by region  
CASE [Parameters].[Region Sales]  
WHEN "EU Sales" THEN [EU Sales]  
WHEN "Global Sales" THEN [Global Sales]  
WHEN "JP Sales" THEN [JP Sales]  
WHEN "NA Sales" THEN [NA Sales]  
WHEN "Other Sales" THEN [Other Sales]  
END
```

Total Sales (All Regions)

```
// Used for calculating other calculated fields- [Region] % of Total  
SUM([NA Sales]) + SUM([EU Sales]) + SUM([JP Sales]) + SUM([Other Sales])
```

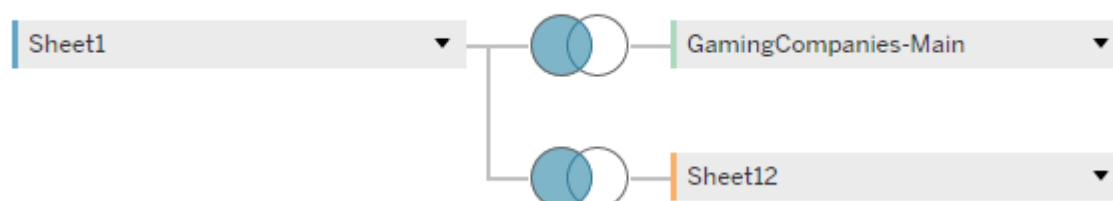
Sales Period <pre>// Checks if the dates are within the defined range IF [Year] >= [Start Date] AND [Year] <= [End Date] THEN 1 ELSE 0 END</pre>

Tableau Worksheets

Sales by Region and Genre	The number of Sales against Year graph
Total Names in the Dataset	KPI for the total games represented inside the dataset
Total Platforms in Dataset	KPI for the total amount of different platforms mentioned in the dataset
Total Publishers in Dataset	KPI for the total amount of different publishers mentioned in the dataset
Global Genres	Acts as a KPI and a filter for genres. It shows the total sales of games by genre and is used if the user wants to filter the data by each genre. Also shows top 5 games in each genre by hovering over the total number of sales.
Tooltip- Top N Games	Used as a tooltip for the global genres to show the top 5 games for each genre
Region Sales %	Pie chart that shows the sales by region and % contribution of each region
Top 5 Publishers	Tree map showing the top 5 publishers based off the video games sales that they were involved in
Top 5 Platforms	Bar graph showing the top 5 platforms that the games were sold on
Top 5 Publisher Earnings 2022	Bar graph showing the top 5 publishers in 2022 based off their annual revenue
Info- Area Graph Info – Genres	An icon that is placed on the dashboard that serves as a tooltip to give extra information

Relationships and Joins

Video Game Sales is made of 3 tables. ⓘ



- “Sheet1” is the VGA dataset or the main one showing all the game titles, publishers, platforms etc. Where “GamingCompanies” and “Sheet12” are joined via “Left Join”, this

was done to ensure none of the data in the VGA dataset becomes null and it's used as a foundation.

- “GamingCompanies” dataset shows the revenue of gaming companies for the year 2022 and is connected to “Sheet1” via ‘Publisher’.
- “Sheet12” is a custom-made table that has ‘Genre’ and URL links to bind the different gaming genres to an icon to improve the visuals in the dashboard. It is connected to “Sheet 1” via ‘Genre’.

Conclusion

The documentation provides the links used for the datasets, a general explanation of the cleaning done on the data with the reasoning behind it and lastly talks about the Tableau calculated fields and joins for the purpose of the user being able to navigate the workbook and understand it.